

Distributed Optimal Task Scheduling of LEO Constellations for Ground Based Assets

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ABSTRACT

The rapid growth of Low Earth Orbit (LEO) constellations has introduced increasingly complex scheduling and resource-allocation challenges across Earth observation, satellite communications, and space situational awareness (SSA) missions. Although orbital motion can be accurately propagated over a planning horizon, satellite-ground access opportunities remain intermittent, and both spacecraft and ground assets are subject to operational limitations such as payload duty cycle, sensing cadence, antenna occupancy, and recovery time. This paper presents a generalized mixed-integer optimization framework for selecting feasible satellite-ground contacts while minimizing the temporal service gaps experienced by mission targets. The formulation represents local operation-cadence constraints and target-side revisit-time objectives using fixed selection and difference matrices, allowing the same scheduling structure to be applied in both satellite-to-ground (SAT-to-GS) and ground-to-satellite (GS-to-SAT) directions. Contact-maximization, square-sum revisit-time minimization, and maximum-revisit-time minimization objectives are considered, together with sequential block-coordinate descent (BCD) and heuristic consensus alternating direction method of multipliers (ADMM) solution strategies. Mission heterogeneity is incorporated by adjusting resource-dependent cadence parameters, target-dependent weighting gains, and the number of parallel payload or ground-resource channels. The proposed framework is demonstrated through Earth-observation image acquisition, 6G satellite gateway scheduling with multi-beam resources, distributed ground-asset scheduling for SSA, and end-to-end Earth-observation image-data download analysis. Numerical results illustrate the trade-offs among resource availability, operational cadence, target priority, and revisit-time performance, demonstrating the applicability of the proposed framework to a broad range of LEO constellation operations.

1. INTRODUCTION

Low Earth Orbit (LEO) constellations provide frequent access opportunities for Earth observation, communications, and space-domain monitoring. Their operational performance, however, depends not only on orbital geometry but also on how the available access opportunities are scheduled. Although future satellite trajectories can be propagated accurately over a planning horizon, satellite-ground visibility remains intermittent, and each space or ground asset is subject to operational constraints such as payload duty cycle, antenna occupancy, recovery time, or sensing cadence. Constellation operation can therefore be viewed as selecting a feasible subset of geometrically available contacts while maintaining a desired temporal service level.

In this context, binary-integer programming based satellite constellation scheduling has consequently received considerable attention. Cho et al. developed an optimization-based scheduling method for agile Earth-observing satellite constellations, demonstrating the applicability of mathematical programming to constellation-level observation planning [1]. More recently, Shtofenmakher and Balakrishnan formulated large-scale SSA tasking as a binary integer program using pre-computed feasibility information [2], while Shimane and Ho employed binary linear programming with a job-shop scheduling structure for single- and multi-sensor ground-based SSA [3]. Beyond centralized optimization, Kim et al. investigated game-theoretic distributed sensor tasking for Earth-observation satellite constellations, highlighting the potential of decentralized decision-making for large-scale scheduling problems [4]. Together, these studies demonstrate the effectiveness of integer-programming and distributed approaches for constellation scheduling, but are generally formulated for mission-specific resource-target configurations.

This paper develops a unified scheduling formulation for both satellite-to-ground (SAT-to-GS) and ground-to-satellite (GS-to-SAT) missions, referred to here as the REDSTONE framework. In SAT-to-GS missions, satellites act as

constrained operating resources and ground locations as service targets, as in Earth observation and satellite communications. In GS-to-SAT missions, ground sensors or stations become the constrained resources and satellites the targets, as in distributed SSA or satellite downlink operations. This symmetry enables a common formulation in which operation-cadence constraints are imposed on the resource side, while revisit or service-gap performance is evaluated on the target side.

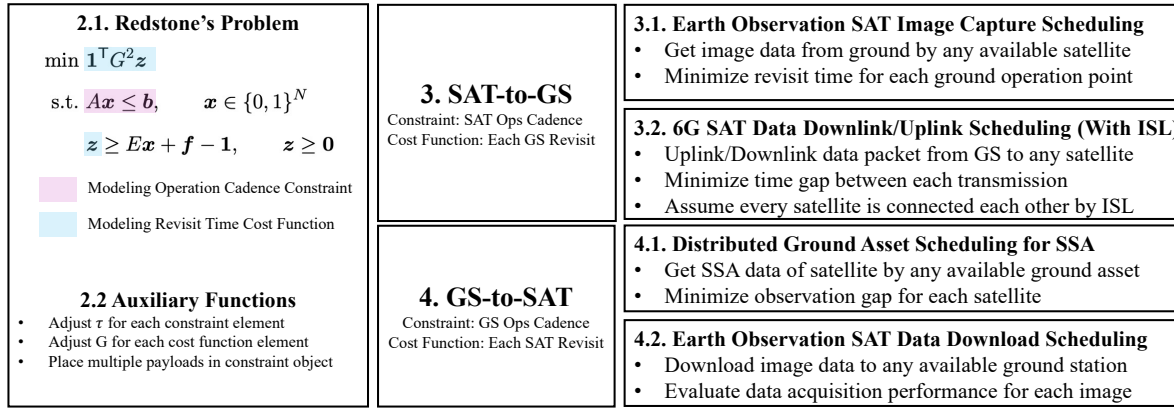


Fig. 1: Overall Configuration: Redstone's problem and SAT-to-GS and GS-to-SAT scheduling applications

Figure 1 summarizes the scope of the paper. The central optimization model framework is first introduced, and framework is subsequently demonstrated in four representative applications: Earth-observation image capture, 6G satellite uplink/downlink scheduling, distributed ground-asset scheduling for SSA, and Earth-observation image-data download. These cases are intended to show that the same contact-selection structure can be reused when the constrained asset and the revisit-time target are interchanged.

2. REDSTONE FRAMEWORK: OPTIMIZATION PROBLEM FOR REVISIT-TIME SCHEDULING

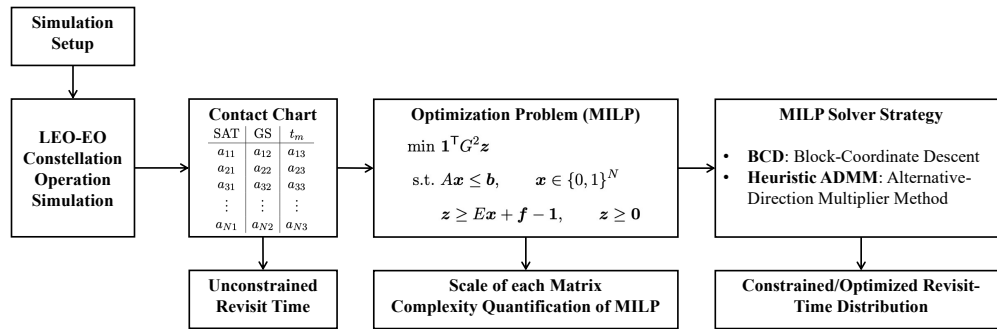


Fig. 2: Overall simulation and optimization framework for LEO constellation operations

The proposed scheduling model separates the problem into three layers: geometric opportunity generation, operational feasibility, and temporal-performance optimization. First, orbit propagation and access analysis produce all candidate satellite-ground contacts over the scheduling horizon. We used MATLAB simulation framework for orbit propagation and generating access chart, which is introduced in previous research [5]. Second, a binary decision vector selects which of these candidate contacts are actually executed while enforcing a minimum operation cadence for each constrained asset. Third, the temporal gaps between consecutive selected contacts are evaluated for each service target and minimized through a global objective. Figure 2 shows this workflow from constellation simulation to the final optimized revisit-time distribution.

Starting from an unconstrained contact chart of geometrically feasible access opportunities, candidate contacts are represented by binary decision variables and selected to satisfy operation-cadence constraints while optimizing revisit-

time performance. Fixed selection and difference matrices are used to construct these temporal relations, yielding a structured formulation that supports contact maximization, squared revisit-time minimization, and maximum-revisit-time minimization. Because the feasibility constraints are naturally separable by operating asset, the centralized problem can also be decomposed into local subproblems and addressed using distributed strategies such as sequential block-coordinate descent (BCD) and parallel alternating direction method of multipliers (ADMM). The detailed matrix construction and mathematical formulation are presented in our previous work [6].

2.1 REDSTONE Framework and Corresponding Distributed Solution approaches

L_1 Problem: Maximize number of contacts $\max \mathbf{1}^T \mathbf{x}$ s.t. $A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$	Constants $A = \begin{bmatrix} E_{S_1,x}^2 & E_{S_1}^1 \\ E_{S_2,x}^2 & E_{S_2}^1 \\ \vdots & \vdots \\ E_{S_p,x}^2 & E_{S_p}^1 \end{bmatrix}, \quad \mathbf{b} = \mathbf{1} + \frac{1}{\tau} \begin{bmatrix} E_{S_1,t}^2 & E_{S_1}^1 \\ E_{S_2,t}^2 & E_{S_2}^1 \\ \vdots & \vdots \\ E_{S_p,t}^2 & E_{S_p}^1 \end{bmatrix} \mathbf{t}$ $E = \begin{bmatrix} E_{G_1,x}^2 & E_{G_1}^1 \\ E_{G_2,x}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,x}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} \mathbf{0}_{1 \times N} \\ \mathbf{I}_{N \times N} \\ \mathbf{0}_{1 \times N} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} E_{G_1,x}^2 & E_{G_1}^1 \\ E_{G_2,x}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,x}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{0}_{N \times 1} \\ 1 \end{bmatrix}$ $G = \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1}^1 \\ E_{G_2,t}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right)$
L_2 Problem: Minimize Square-sum Revisit Time $\min \mathbf{1}^T G^2 \mathbf{z}$ s.t. $A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$ $\mathbf{z} \geq E\mathbf{x} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z} \geq \mathbf{0}$ $\mathbf{z} \in R^M, M = \sum_{j=1}^q \frac{(G_j + 2)(G_j + 1)}{2}$	
L_∞ Problem: Minimize Maximum Revisit Time $\min R$ s.t. $A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$ $R\mathbf{1} \geq G(E\mathbf{x} + \mathbf{f} - \mathbf{1})$	

Fig. 3: **REDSTONE (REvisit-time Determination and Satellite Tasking Optimization) Equations) Framework**

In this section, we introduce REDSTONE framework, which provides a unified mathematical framework for determining, standardizing, and optimizing revisit-time performance from a discrete contact chart. Its central idea is to separate the scheduling problem into resource-side operational feasibility and target-side temporal service performance.

In this framework, we introduce three types of objective functions. The L_1 objective function ($\mathbf{1}^T \mathbf{x}$) maximizes the number of selected contacts, which is useful for producing a high-utilization schedule but does not directly balance the temporal gaps among targets. The L_2 objective function ($\mathbf{1}^T G^2 \mathbf{z}$) minimizes the sum-square of activated revisit times. Finally, the L_∞ objective function (R) minimizes the longest revisit interval. These objectives allow the same feasibility model to emphasize utilization, average/fair temporal performance, or worst-case service guarantees.

The rows of $A\mathbf{x} \leq \mathbf{b}$ can be partitioned by operating asset, so the binary variables associated with one satellite appear only in that satellite's local feasibility block. In contrast, the revisit objective couples the agents because contacts from different satellites can serve the same ground point. Thus, feasibility is locally separable while performance is globally coupled. The derivation process and detailed theoretical insights are described in our previous research [6].

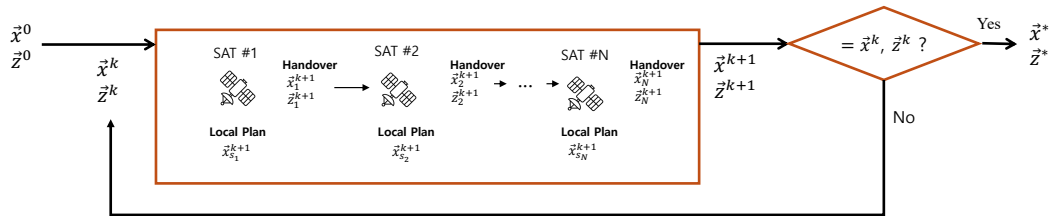


Fig. 4: Sequential block-coordinate descent (BCD) framework for satellite-wise distributed task scheduling.

Figure 4 describes the framework BCD approach. The global decision vector is partitioned into satellite-wise blocks. At iteration k , satellite i re-optimizes only its local decision block while the current decisions of all other satellites

are held fixed. The accepted local update is inserted into the global schedule and passed to the next satellite. One complete sweep therefore consists of a sequence of local MILP updates. The procedure is repeated until a complete sweep produces no further improvement in the global objective. This sequential architecture is attractive when reliable objective reduction is more important than parallel execution, because each local solution is evaluated against the current global schedule before it is handed over to the next agent.

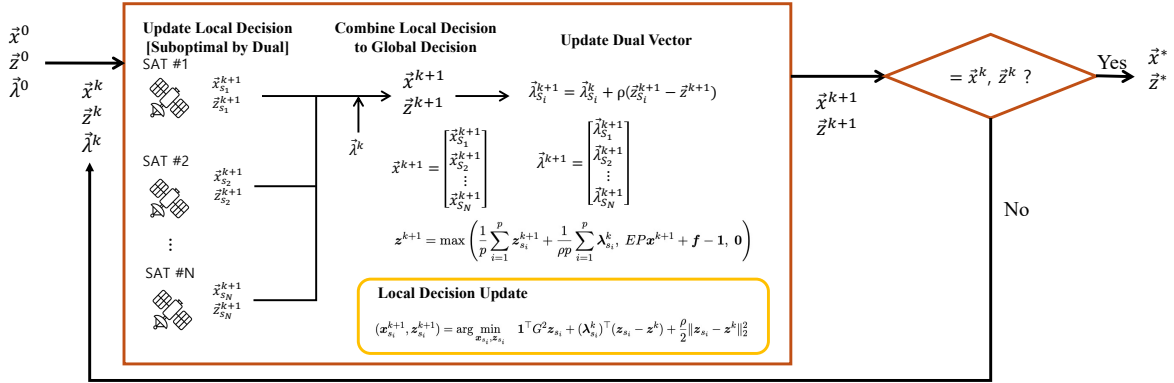


Fig. 5: Heuristic ADMM framework with satellite-wise local decision updates, global consensus updates, and dual-variable updates.

To permit parallel local updates, the heuristic ADMM strategy in Figure 5 gives each satellite a local copy of the revisit variable and augments the local objective with a consensus penalty. All satellites can then solve their local mixed-integer subproblems simultaneously. The local solutions are combined to update a global revisit representation, after which dual variables penalize disagreement between the local and global copies. The penalty parameter ρ controls the strength of this agreement mechanism: a small value permits greater local freedom but may produce slow or oscillatory coordination, whereas a large value drives rapid consensus but can bias the mixed-integer solution away from the best objective value.

Because the local subproblems retain binary variables, the ADMM implementation is used here as a heuristic distributed method rather than as a convex ADMM algorithm with a global convergence guarantee. In the previous research [6], we showed the expected trade between the two distributed strategies: BCD generally achieved lower final objective values, whereas ADMM substantially reduced the number of coordination iterations through parallel updates. Since we didn't utilize parallel processing in this paper's simulations, and ADMM required to decide adequate hyperparameter ρ , we used BCD for primary solving method.

2.1.1 Selected Methodology: Solving L_2 Problem and using BCD Solution Approach

The role of the objective function is first examined using a controlled regional Earth-observation example. Table 1 defines a 48-satellite, 500-km circular constellation and 54 lattice points surrounding the Korean Peninsula.

Table 1: Lattice point mission specification.

Parameter	Lattice Point Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 1 day, Timestep: 5 seconds
Ground Points	54 lattice points around Korea Peninsula: (33°N–43°N, 119°E–135°E)
Orbit: Altitude, Shape	500 km, Circular Orbit
Orbit: Inclination Angle	98°
Payload FOV	60°
Specific Mission	Simple Conic Payload FOV, Slew roll angle for capturing
Constellation Config.	48 SATs: 12 Orbit Planes, 4 SATs/plane
No. of Total Missions	1500

The experiment is deliberately constructed as a scheduling benchmark: the unconstrained contact chart is generated

first, and the same set of candidate contacts is then re-optimized under a 30 s operation-cadence requirement. This makes it possible to isolate the effect of the objective and the distributed solver without changing the orbital geometry.

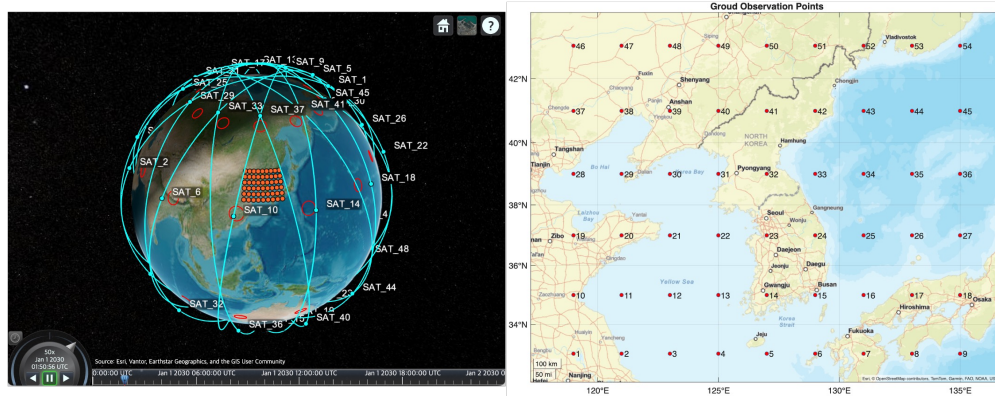


Fig. 6: Simulation visualization for lattice point mission

Figure 6 shows both the constellation geometry and the regional lattice used for this benchmark. A conical sensor footprint and allowable roll pointing generate the feasible observation opportunities. The resulting candidate events are time ordered and supplied directly to the contact-chart optimization; therefore, all solutions compared below operate on the same geometric access set.

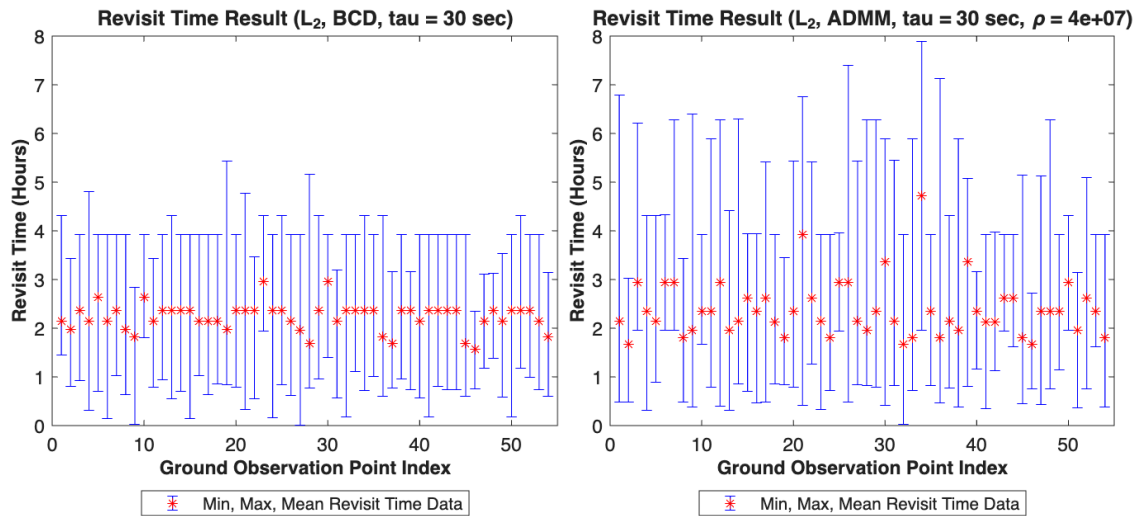


Fig. 7: Ground-point revisit-time distributions using L_2 -based BCD and heuristic ADMM with $\tau = 30$ s.

Figure 7 compares the target-wise minimum, maximum, and mean revisit times obtained by the L_2 BCD and heuristic ADMM schedules. Both methods distribute observations across the full lattice rather than simply concentrating on the most frequently accessible points. The BCD solution exhibits a comparatively compact distribution of the mean revisit times and smaller worst-case excursions for most locations. The ADMM result preserves the same overall pattern but shows a wider spread for several ground points, consistent with the loss of solution quality that can accompany aggressive consensus enforcement in a mixed-integer problem. The result supports the use of BCD when the primary objective is balanced revisit performance rather than capability of parallel processing.

The contact-maximizing L_1 solution in Figure 8 illustrates why contact count alone is not an adequate surrogate for revisit performance. A schedule can select many contacts while still leaving long temporal gaps at particular ground points. When the subsequent L_∞ BCD optimization reduces the worst-case spread but remains relatively conservative, with many ground points pushed toward a common high revisit-time level.

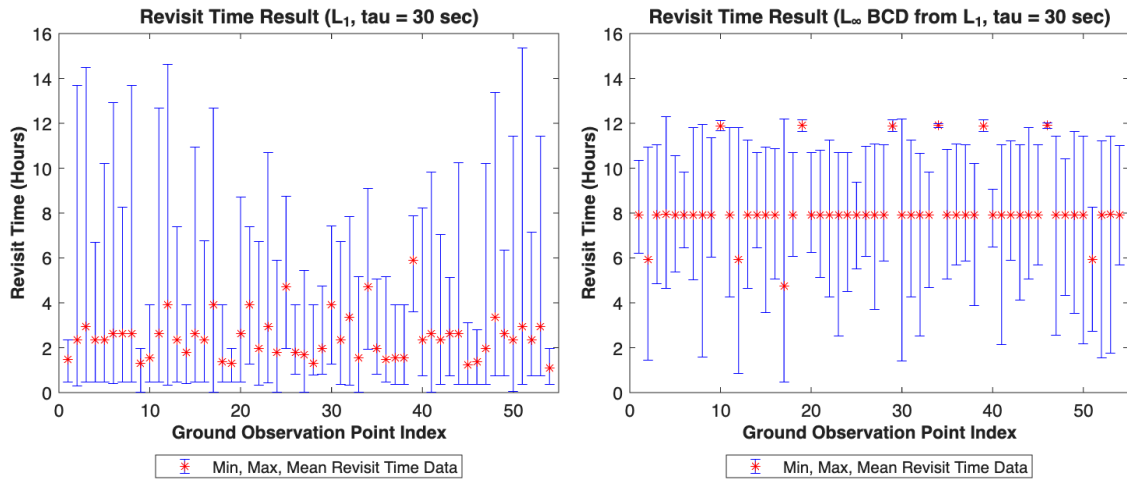


Fig. 8: Ground-point revisit-time performance obtained from the L_1 solution and the subsequent L_∞ BCD optimization.

In contrast, Figure 9 starts the L_∞ refinement from the L_2 BCD solution. Since the L_2 objective already penalizes large temporal gaps and encourages fairness across ground points, the initial schedule is much closer to a balanced revisit configuration. The resulting worst-case refinement therefore retains substantially smaller mean and maximum revisit levels than the solution initialized from the L_1 schedule. However, overall mean revisit time is higher in L_∞ refinement. Therefore, for the remainder of the paper, the L_2 BCD formulation is consequently used as the main scheduling tool because it provides a direct and interpretable compromise between overall revisit reduction and fairness.

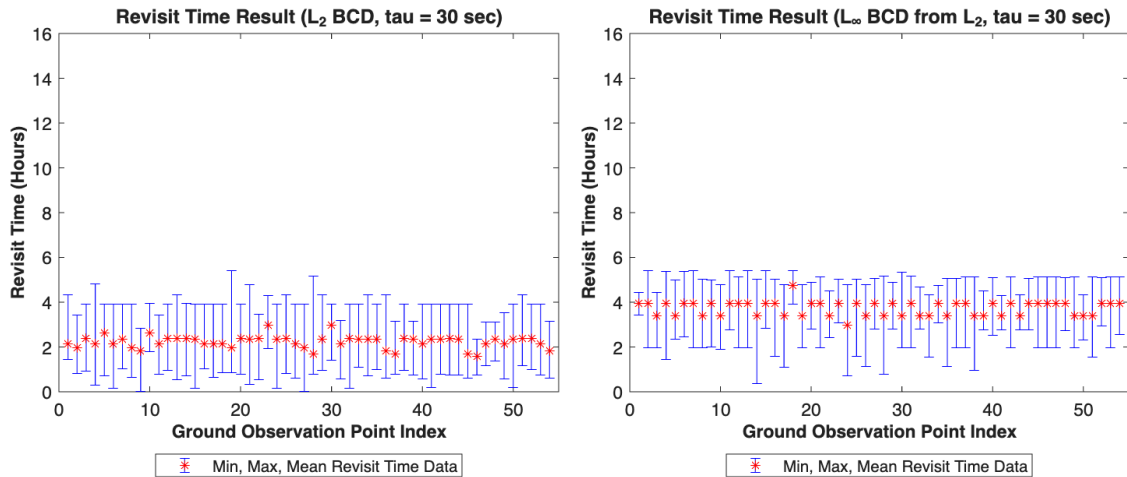


Fig. 9: Ground-point revisit-time performance obtained from L_2 BCD and the subsequent L_∞ BCD optimization.

2.2 Adjusting τ and G for Customizing the Constraint and Objective to each Object

A major advantage of the selection-matrix representation is that mission heterogeneity can be introduced without changing the binary decision structure. Two particularly useful design parameters are the operation-cadence vector and the revisit-weighting matrix. Rather than assigning one common cadence τ to every constrained asset, each satellite or ground asset can be assigned its own value τ_i . Likewise, the diagonal entries of G can be multiplied by target-dependent gains to prioritize selected ground points or satellites.

Figure 10 replaces the scalar factor $1/\tau$ in the global right-hand side with a block-diagonal scaling based on $1/\tau_i$. This allows assets with different payload recovery times, duty-cycle limitations, or service capabilities to coexist in the same optimization.

Adjusting τ to customize the operation cadence constraint

$$\mathbf{b} = \mathbf{1} + \frac{1}{\tau} \begin{bmatrix} E_{S_1,t}^2 & E_{S_1,t}^1 \\ E_{S_2,t}^2 & E_{S_2,t}^1 \\ \vdots & \vdots \\ E_{S_p,t}^2 & E_{S_p,t}^1 \end{bmatrix} \mathbf{t} \implies \mathbf{b}' = \mathbf{1} + \begin{bmatrix} \frac{1}{\tau_{S_1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{\tau_{S_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\tau_{S_p}} \end{bmatrix} \begin{bmatrix} E_{S_1,t}^2 & E_{S_1,t}^1 \\ E_{S_2,t}^2 & E_{S_2,t}^1 \\ \vdots & \vdots \\ E_{S_p,t}^2 & E_{S_p,t}^1 \end{bmatrix} \mathbf{t}$$

Fig. 10: Customization of the operation-cadence constraint using satellite-dependent cadence parameters τ_{S_i}

Figure 11 applies an analogous target-side scaling. If a diagonal gain $k_{G_j} > 1$ is assigned to a selected ground point, temporal gaps associated with that location become more expensive and the optimizer is encouraged to allocate a larger share of the available contact opportunities to it. The following case studies use these two mechanisms, together with replicated resource channels, to model a variety of mission architectures.

Adjusting G by gain diagonal matrix to customize the objective function

$$G = \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1,t}^1 \\ E_{G_2,t}^2 & E_{G_2,t}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q,t}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right) \implies G' = \begin{bmatrix} k_{G_1} & 0 & \cdots & 0 \\ 0 & k_{G_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & k_{G_q} \end{bmatrix} \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1,t}^1 \\ E_{G_2,t}^2 & E_{G_2,t}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q,t}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right)$$

Fig. 11: Customization of revisit-time objective using diagonal gain matrix to assign different priorities to each GS

3. SAT TO GS: CONSTRAINT TO EACH SAT / OPTIMIZE REVISIT TIME OF EACH GS

SAT-to-GS Constraint: SAT Ops Cadence Cost Function: Each GS Revisit	Earth Observation SAT Image Capture Scheduling - Get image data from ground by any available satellite - Minimize revisit time for each ground operation point	- Worldwide Distributed Observation - Customize τ (Operation Cadence) for each SAT / G to Prioritize Ground point
	6G SAT Data Downlink/Uplink Scheduling (With ISL) - Uplink/Downlink data packet from GS to any satellite - Minimize time gap between each transmission - Assume every satellite is connected each other by ISL	- Regional Lattice Points Communication - Place Multiple Communication Payloads in single SAT (Multi-Beamforming)

Fig. 12: SAT-to-GS scheduling applications for Earth-observation image acquisition and 6G uplink/downlink

In the SAT-to-GS orientation, each satellite is treated as the constrained operating asset. Its candidate tasks may correspond to imaging, transmission, or another payload operation, and the satellite-specific cadence constraint limits how rapidly those tasks can be executed. Ground locations are the service targets, so the objective evaluates the time between successive services received at each ground point. Figure 12 highlights two applications considered in this paper. The first schedules image acquisition opportunities for a distributed set of Earth-observation points. The second schedules satellite-to-gateway communication opportunities in a 6G non-terrestrial network (NTN) architecture.

3.1 Earth-Observation Satellite: Image capture scheduling

For Earth-observation image capture, a candidate task is created whenever a ground observation point lies within the available sensor-access region of a satellite. Figure 13 distinguishes the instantaneous conical field of view from the discrete image-capture locations used by the scheduler. The conic footprint determines geometric accessibility, while roll steering allows the payload line of sight to be displaced within the available cross-track region. Once these candidate imaging events are generated, the optimizer chooses a subset that respects the satellite operation cadence and minimizes the revisit-time distribution of the ground observation points.

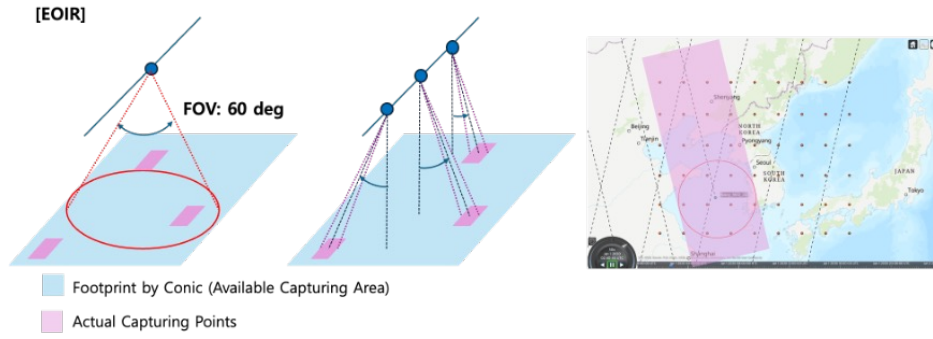


Fig. 13: Earth-observation imaging geometry illustrating the sensor FOV, accessible footprints

Table 2: Earth Observation mission specification.

Parameter	Earth Observation Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 7 days, Timestep: 5 seconds
Ground Points	31 globally distributed ground observation points
Orbit: Altitude, Shape	500 km, Circular Orbit
Orbit: Inclination Angle	98°
Payload FOV	60°
Specific Mission	Simple Conic Payload FOV, Slew roll angle for capturing
Constellation Config.	48 SATs: 12 Orbit Planes, 4 SATs/plane
No. of Total Missions	1500 → Actual Duration: 24 hours

Table 2 defines the worldwide Earth-observation case. The simulation uses a 500-km circular orbit with a 98° inclination and a 48-satellite constellation arranged in 12 orbital planes. The sensor field of view is 60°, and the resulting worldwide candidate contacts are collected over the specified horizon. Figure 14 shows the orbital geometry together with the geographic distribution of the ground observation points used to evaluate temporal coverage.

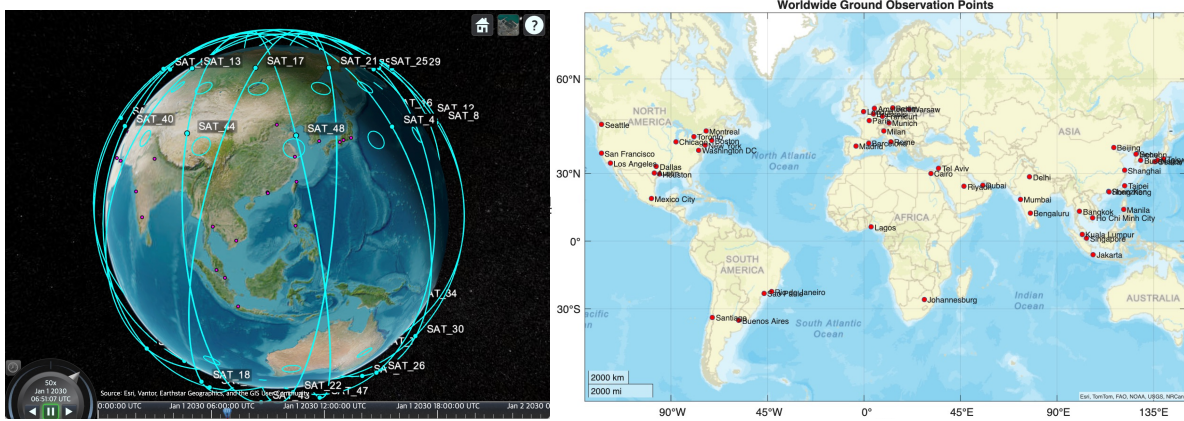


Fig. 14: LEO constellation simulation and worldwide distribution of ground observation points.

The unconstrained schedule serves as the geometric upper-bound case in which every candidate imaging opportunity is accepted. Operationally constrained cases are then generated by increasing τ . This parameter sweep quantifies the direct trade between satellite workload and service regularity: a larger cadence forces more candidate contacts to be discarded, which necessarily increases the temporal gaps experienced at the ground points.

3.1.1 Revisit-Time Result by Adjusting τ

Table 3 summarizes this trade over cadence values from 1000 to 4000 s. The number of selected observations decreases monotonically as the cadence is tightened, while both the square-gap objective and the revisit-time statistics increase. This behavior is expected: the orbit geometry does not change, but the scheduler is allowed to exploit a progressively smaller fraction of the available access opportunities.

Table 3: Revisit time results with respect to operation cadence τ .

τ [s]	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean Revisit [min]	Max Revisit [min]
Unconstrained	1500	1.88×10^{10}	50.5	157.0
1000	718	3.69×10^{10}	101.3	275.7
2000	665	3.86×10^{10}	108.7	306.8
3000	559	4.68×10^{10}	127.8	427.2
4000	520	5.09×10^{10}	136.3	476.1

Numerically, the unconstrained case selects all 1500 candidate events and yields a mean revisit time of 50.5 min with a maximum of 157.0 min. At $\tau = 1000$ s, only 718 contacts remain and the mean revisit increases to 101.3 min. At $\tau = 4000$ s, the selected set is reduced to 520 contacts, producing a mean revisit of 136.3 min and a maximum of 476.1 min. Thus, the cadence parameter provides a direct operational knob for trading payload usage against temporal coverage performance.

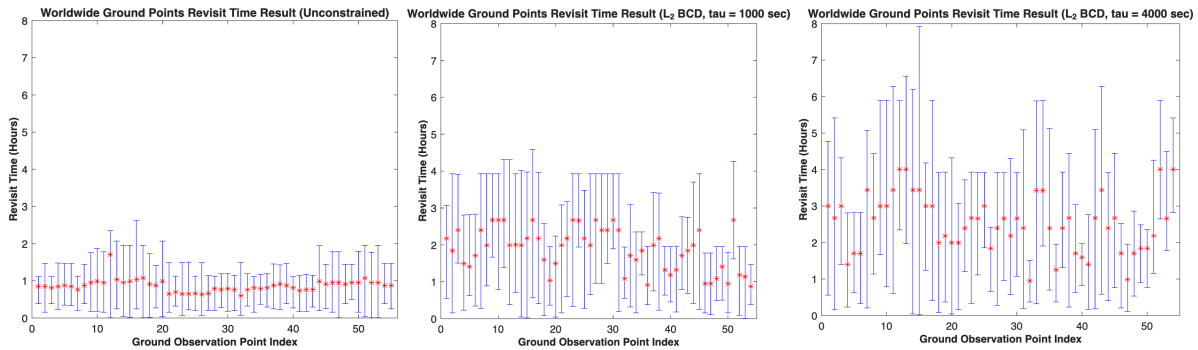


Fig. 15: Worldwide ground-point revisit-time distributions for the unconstrained case and L_2 BCD solutions with different operation-cadence constraints.

The same trend is visible spatially in Figure 15. The unconstrained solution has a relatively compact revisit distribution. As the cadence grows, the mean values rise and the min–max bars expand for ground points with fewer interchangeable overflight opportunities. The nonuniform degradation is important: a single global cadence does not penalize every location equally because the underlying access geometry differs by latitude and constellation phasing.

3.1.2 Adjusting G to Prioritize a Selected Ground Observation Point

The target-weighting mechanism is next demonstrated by increasing the gain associated with GS15, located at Jakarta, Indonesia. All other model parameters are held fixed. Increasing the priority weight k makes revisit gaps at GS15 progressively more expensive in the L_2 objective, encouraging the optimizer to reallocate contacts toward that location without explicitly imposing a hard revisit-time constraint.

Table 4 shows that the number of selected contacts assigned to GS15 increases from 5 at $k = 1$ to 17 at $k = 10$. Over the same range, the GS15 mean revisit time decreases from 241.1 min to 80.4 min, and its maximum revisit decreases from 476.2 min to 121.5 min. Meanwhile, the constellation-wide mean revisit remains close to 136 min. This demonstrates that the diagonal weighting can strongly improve service to a designated high-priority location while preserving the same basic global feasibility model.

Table 4: Overall and GS15 (Jakarta, Indonesia) revisit time results with respect to the priority weight k .

k	Overall Revisit Time				GS15 (Jakarta, Indonesia) Revisit Time			
	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean [min]	Max [min]	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean [min]	Max [min]
1	520	5.09×10^{10}	136.3	476.2	5	1.58×10^9	241.1	476.2
2	521	5.09×10^{10}	136.1	512.5	10	8.16×10^8	131.5	239.6
5	522	5.21×10^{10}	136.1	597.7	15	5.80×10^8	90.4	121.5
10	522	5.25×10^{10}	136.1	597.7	17	5.62×10^8	80.4	121.5
Unconstrained	1500	1.88×10^{10}	50.6	157.0	22	5.49×10^8	62.9	118.5

Figure 16 shows the redistribution of revisit performance over all ground points, while Figure 17 isolates the temporal schedule of GS15. The unweighted constrained case contains several long service gaps. As the priority is increased, additional contacts are inserted into those gaps and the time history becomes substantially more regular. The result illustrates how a soft priority can be represented through $\mathbf{G}$ without changing the orbit propagation or introducing a separate mission-specific integer constraint.

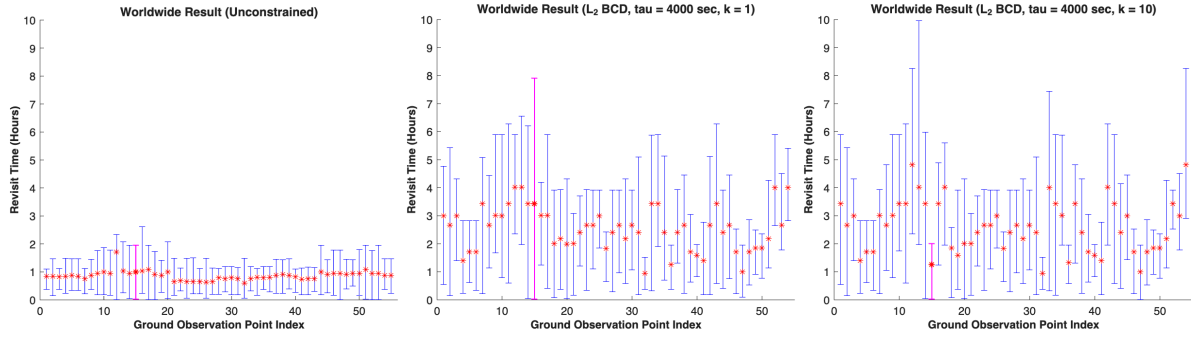


Fig. 16: Effect of ground-point priority weighting on worldwide revisit-time distributions under the L_2 BCD solution.

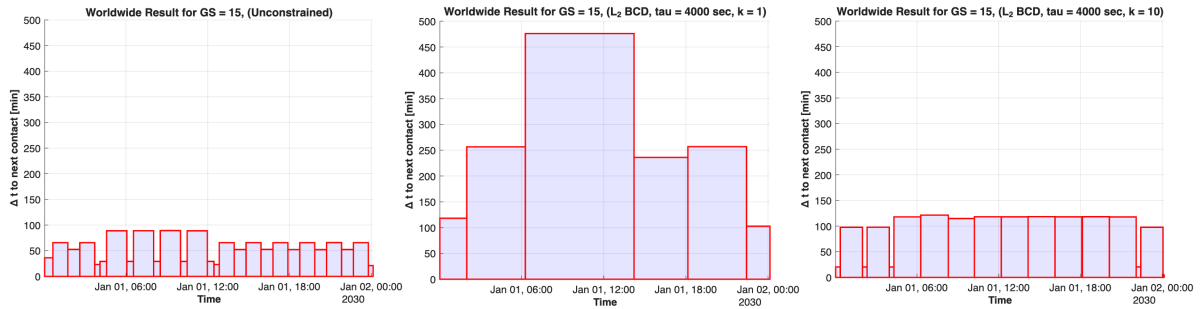


Fig. 17: Contact-interval history of a representative prioritized ground point.

3.2 6G Telecom Satellite Data Downlink/Uplink Scheduling with ISL

The same SAT-to-GS scheduling structure can represent a simplified 6G NTN service problem. Here, the ground points correspond to gateway cells and a candidate contact represents an available satellite–gateway communication opportunity. Inter-satellite links (ISLs) are assumed to provide connectivity among the satellites, so the scheduling layer focuses on when each gateway can be served by the constellation rather than on the detailed routing of a packet through the space network. Detailed routing strategy using MDP is described in our previous research [7]. The objective is therefore expressed as the communication-period gap at each gateway, while per-satellite cadence limits the rate at which a spacecraft can commit communication resources.

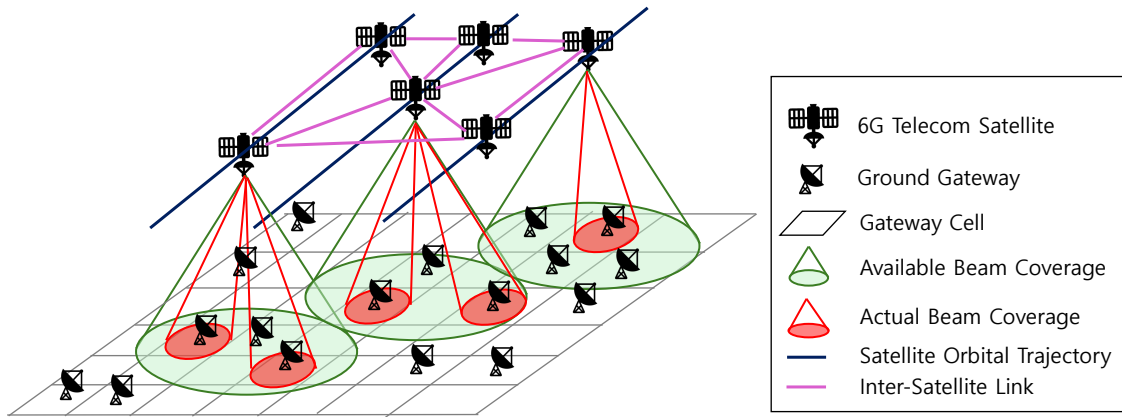


Fig. 18: Graphical representation of an observation or coverage footprint used in the scheduling framework.

Figure 18 depicts the abstraction used for multi-beam scheduling. The green region denotes the geometrically available beam footprint and the red footprint denotes an actually selected service assignment. Multiple payload channels on a satellite are modeled as multiple available communication resources, allowing the optimization to represent several concurrently serviceable beams without changing the ground-point revisit objective.

Table 5: 6G telecom constellation mission specification.

Parameter	6G Telecom Constellation Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 1 hour, Timestep: 5 seconds
Ground Points	56 hexagonal cell grid points in South Korea
Orbit: Altitude, Shape	1200 km, Circular Orbit
Orbit: Inclination Angle	38°
Payload FOV	60°
Specific Mission	Multiple communications payloads (1, 2, 3, 4) per satellite
Constellation Config.	128 SATs: 16 Orbit Planes, 8 SATs/plane
satellite operation cadence (τ)	20s

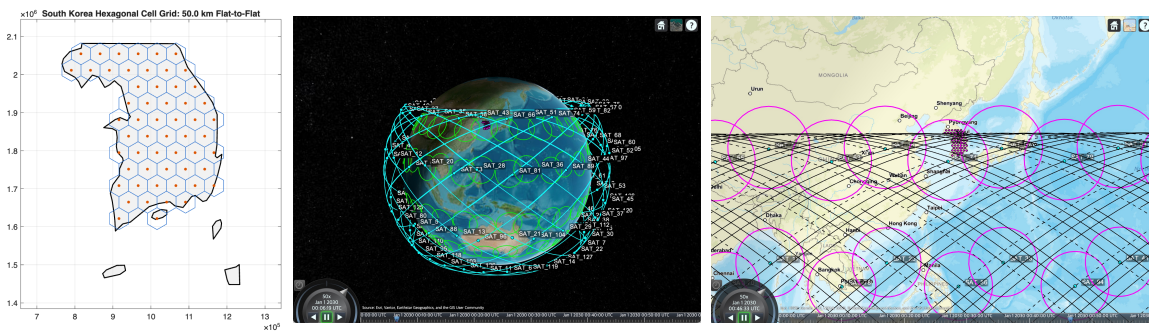


Fig. 19: 6G NTN constellation simulation setup including the hexagonal ground-gateway grid, LEO constellation, and multi-beam communication geometry.

Table 5 and Figure 19 define the communications case. Fifty-six gateway cells are distributed over South Korea, and 128 satellites in 16 orbital planes provide the candidate service opportunities during the one-hour simulation. The ground grid is intentionally dense relative to the regional footprint, making simultaneous demand for satellite resources a prominent scheduling feature.

3.2.1 Modeling Multi-Beam Operation with Multiple Payload Resources

Multi-beam capability is represented by increasing the number of communication payload resources associated with each satellite from one to four. In the scheduling model, these replicated resources increase the number of gateway assignments that can be accepted while preserving the local cadence of each resource. The experiment therefore quantifies the operational value of additional beam capacity at the contact-scheduling level.

Table 6: Revisit time results with respect to the number of payloads per satellite.

No. of Payloads	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean Revisit [min]	Max Revisit [min]
1	56	4.49×10^8	29.0	57.9
2	110	2.90×10^8	19.6	40.4
3	162	2.14×10^8	14.9	31.3
4	209	1.72×10^8	12.2	26.3
Unconstrained	920	6.62×10^6	1.91	3.33

As shown in Table 6, increasing the number of payload resources produces a nearly monotonic improvement in gateway service. With one payload, 56 communication events are selected and the mean communication period is 29.0 min. With four payloads, 209 events are selected and the mean period falls to 12.2 min, while the maximum decreases from 57.9 min to 26.3 min. The unconstrained contact chart remains substantially better, with a 1.91 min mean period, indicating that payload multiplicity recovers part, but not all, of the service flexibility lost to operational constraints.

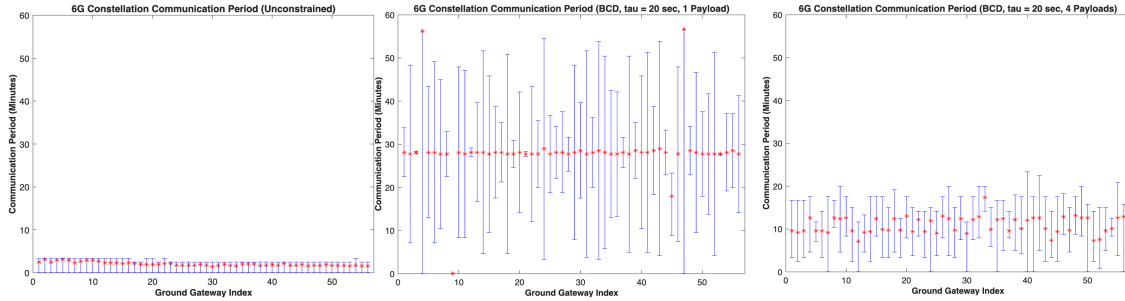


Fig. 20: Ground-gateway communication-period distributions with one and four communication payloads.

Figure 20 makes this improvement visible across the entire gateway grid. A single communication resource leads to long and widely varying service periods. Four resources compress both the mean values and the min–max spread, moving the constrained result toward the unconstrained geometric limit. This case should be interpreted as a constellation-operations model of multi-beam resource availability; detailed waveform design, interference management, frequency assignment, queueing, and link-capacity constraints are outside the present contact-scheduling abstraction and can be added as additional feasibility layers in future work.

4. GS TO SAT: CONSTRAINT TO EACH GS / OPTIMIZE REVISIT TIME OF EACH SAT

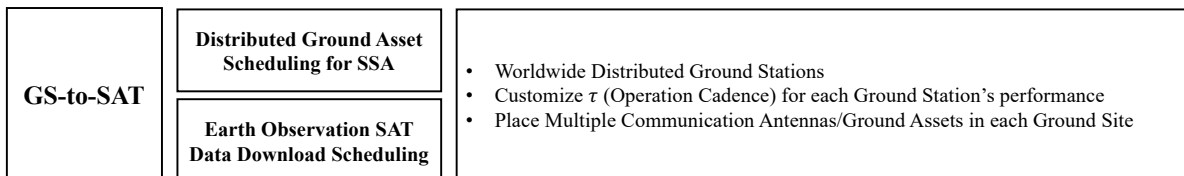


Fig. 21: GS-to-SAT scheduling applications for distributed ground-asset scheduling for space situational awareness and Earth-observation satellite data download scheduling.

The GS-to-SAT orientation reverses the roles of resource and target. Ground stations, antennas, or sensors are now the constrained operating assets, and the temporal performance metric is evaluated for each satellite. This representation is appropriate when the principal operational bottleneck is on the ground side, such as limited tracking-sensor cadence, antenna occupancy, or the number of independent ground terminals. Figure 21 summarizes the two examples considered here: distributed SSA tracking and Earth-observation image-data download.

4.1 Distributed Ground Asset Scheduling for Space Situational Awareness

For the SSA application, each candidate event is a line-of-sight observation opportunity between a ground sensor and a LEO satellite. The local cadence constraint is imposed on each ground asset, reflecting finite sensor occupancy and the need to transition or recover between observations. The target-side objective evaluates the observation gap of every satellite, so a good schedule must distribute the available ground-sensor time across the constellation rather than repeatedly observing only the easiest objects. Figure 22 illustrates this resource-allocation view, including the possibility of deploying additional sensors at a high-demand location.

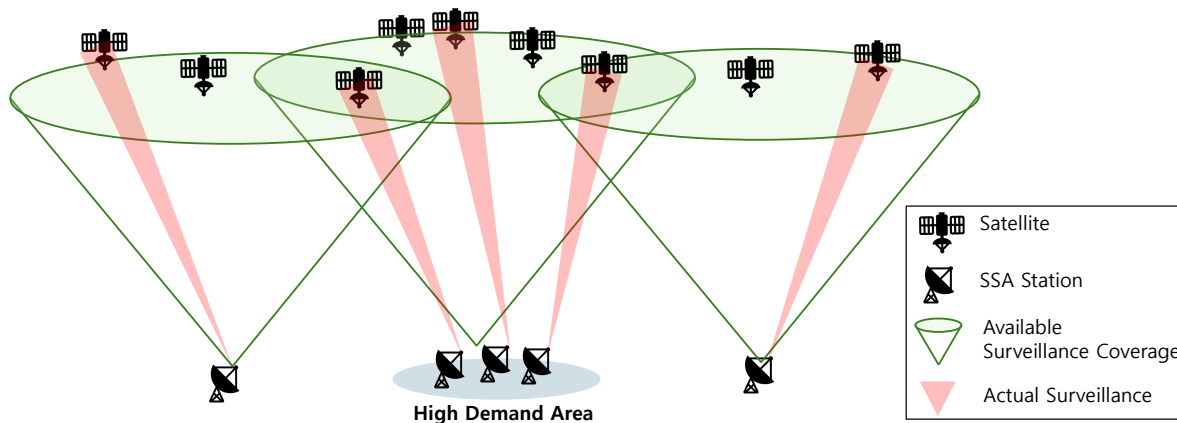


Fig. 22: Graphical representation used in the GS-to-SAT scheduling framework.

The baseline network in Figure 23 contains geographically distributed sites, including Svalbard, Tromso, Daejeon, Awarua, McMurdo, TrollSat, Punta Arenas, Fairbanks, and Inuvik. Table 7 uses a one-day scheduling horizon and the same 48-satellite LEO constellation architecture as the Earth-observation case. Svalbard is treated as the representative high-demand site for the subsequent multi-asset experiment because of its frequent access to high-inclination LEO trajectories.

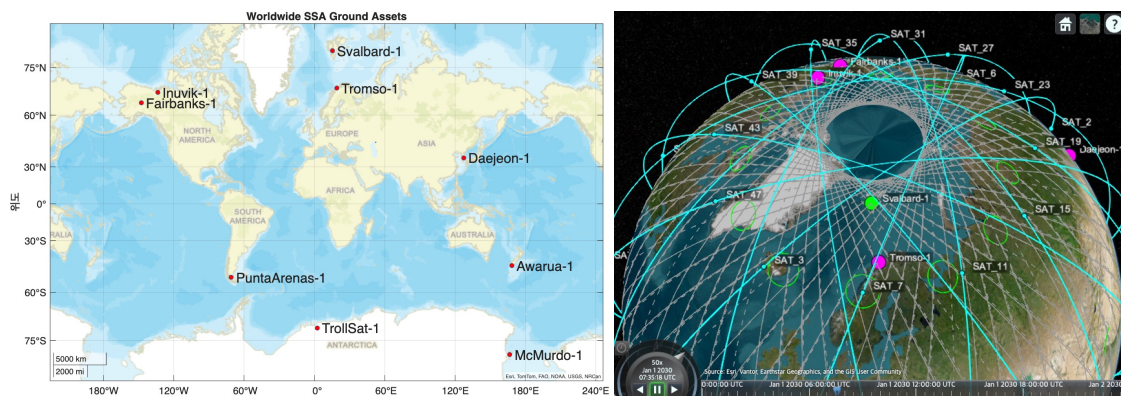


Fig. 23: Worldwide distribution of ground assets and LEO constellation geometry for distributed SSA scheduling.

Table 7: Distributed ground asset scheduling mission specification for space situational awareness.

Parameter	Distributed Ground Asset Scheduling Mission Specification
Time Horizon	Starting Time: Jan. 1, 2030, Duration: 1 day, Timestep: 5 seconds
Ground Assets	9 selected ground assets distributed worldwide
Orbit: Altitude, Shape	500 km, Circular Orbit
Orbit: Inclination Angle	98°
Minimum Elevation	57.3°
Specific Mission	Multiple ground assets deployed in a high-demand area (Svalbard)
Constellation Config.	48 SATs: 12 Orbit Planes, 4 SATs/plane

The first SSA experiment varies a common ground-asset cadence τ . As in the SAT-to-GS case, increasing τ reduces the number of observations that can be accepted. Here, however, the resulting penalty appears as a larger satellite observation gap because the satellites are the service targets.

Table 8: Revisit time results with respect to operation cadence τ for distributed ground asset scheduling.

τ [s]	$\mathbf{1}^T \mathbf{x}$	$\mathbf{1}^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean Revisit [min]	Max Revisit [min]
Unconstrained	671	3.86×10^{10}	95.4	427.8
500	468	4.60×10^{10}	134.0	428.3
1000	398	5.05×10^{10}	155.0	429.7
1500	275	6.97×10^{10}	213.6	558.3
2000	246	7.55×10^{10}	235.5	649.5

Table 8 shows that the unconstrained network selects 671 observations and produces a mean satellite observation gap of 95.4 min. At $\tau = 500$ s, the selected set decreases to 468 observations and the mean gap increases to 134.0 min. At $\tau = 2000$ s, only 246 observations remain, with a mean gap of 235.5 min and a maximum of 649.5 min. The rapid increase at large cadence values indicates that ground-resource occupancy can become a dominant limitation even when the underlying constellation remains continuously predictable.

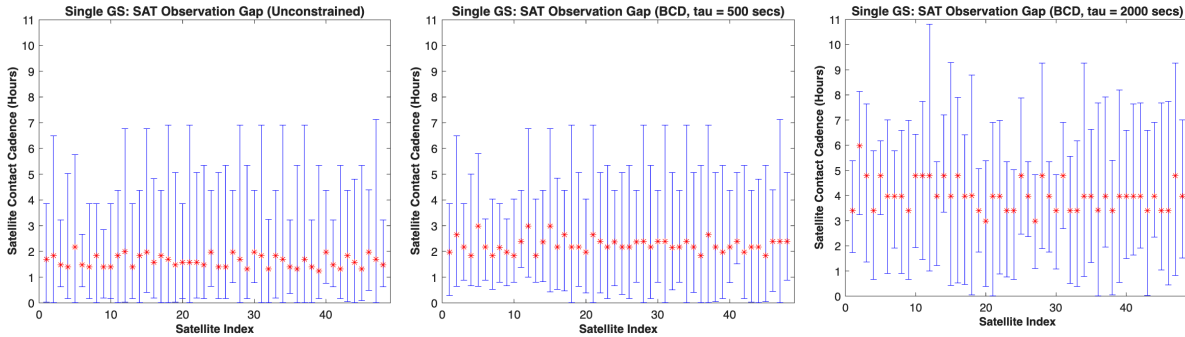


Fig. 24: Satellite observation-gap distributions for a single ground station under the unconstrained case and BCD solutions with operation-cadence constraints of $\tau = 500$ s and $\tau = 2000$ s.

Figure 24 shows the corresponding satellite-wise distributions for a representative ground station. The unconstrained and moderate-cadence cases preserve relatively frequent observations for most satellites, whereas the 2000 s case shifts the distribution upward and increases the spread. This illustrates the same general resource-versus-revisit trade identified earlier, but with the ground network rather than the spacecraft acting as the limiting resource.

4.1.1 Placing Multiple Ground Assets at the High-Demand Svalbard Site

We can improve temporal coverage by increasing number of independent resources at a heavily utilized site. Multiple colocated Svalbard assets are modeled as separate ground-resource channels, each with its own cadence constraint.

Although the assets share essentially the same visibility geometry, they can accept different satellite contacts in parallel and therefore increase the aggregate service capacity of the site.

Table 9 quantifies the network-level benefit. Increasing the number of Svalbard assets from one to four raises the total selected observations from 275 to 370 and reduces the overall mean satellite revisit from 213.6 min to 166.3 min. The overall objective function also decreases as additional assets are introduced. At the individual-resource level, each Svalbard asset maintains a mean interval of approximately 41–42 min in the multi-asset cases, showing that the network improvement is obtained by distributing the workload among independent channels rather than by violating the cadence of a single sensor.

Table 9: Overall and individual ground-station revisit time results with respect to the number of ground stations.

Number of GS at Svalbard	Overall Revisit Time				Svalbard GS Index	Individual GS Revisit Time			
	$1^T \mathbf{x}$	$1^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean [min]	Max [min]		$1^T \mathbf{x}$	$1^T \mathbf{G}^2 \mathbf{z}$ [s ²]	Mean [min]	Max [min]
1	275	6.97×10^{10}	213.6	558.3	1	33	2.37×10^8	42.3	71.5
2	303	6.43×10^{10}	196.6	558.3	2	33	2.39×10^8	42.3	70.3
3	341	5.99×10^{10}	177.4	556.6	3	33	2.33×10^8	42.3	70.4
4	370	5.83×10^{10}	166.3	556.6	4	34	2.27×10^8	41.1	66.8
—	—	—	—	—	Single GS	38	2.05×10^8	36.9	66.9

Figure 25 verifies this behavior in the time domain for the four colocated Svalbard assets at $\tau = 1500$ s. The selected-contact intervals of the individual resources remain separated by the prescribed cadence scale, while their schedules are interleaved across the day. In practical terms, the model treats additional antennas or sensors as a direct increase in parallel service capacity and can therefore be used to study whether temporal-performance requirements are better met by changing scheduling policy or by adding ground infrastructure.

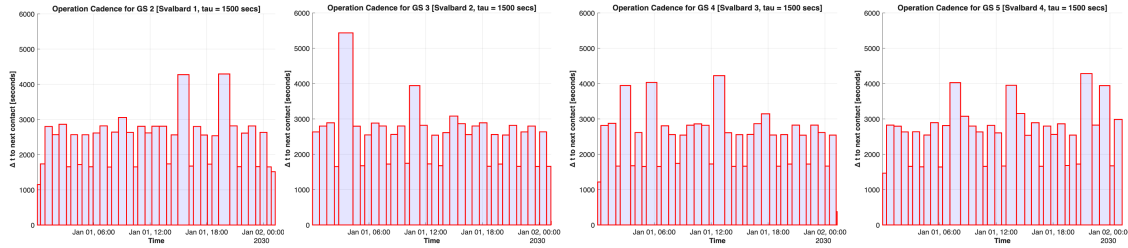


Fig. 25: Operation cadence of multiple ground assets colocated at Svalbard with $\tau = 1500$ s.

4.2 Earth Observation Satellite: Image Data Download Scheduling

Revisit-time optimization of image acquisition does not by itself guarantee timely delivery of the collected data. An image becomes operationally useful only after it can be transferred from the spacecraft to an available ground station. The final case therefore connects the SAT-to-GS image-acquisition process to a GS-to-SAT ground-network process and evaluates the elapsed time between imaging and downlink. Figure 26 illustrates this end-to-end sequence from worldwide observation points, through the LEO constellation, to the distributed ground-station network.

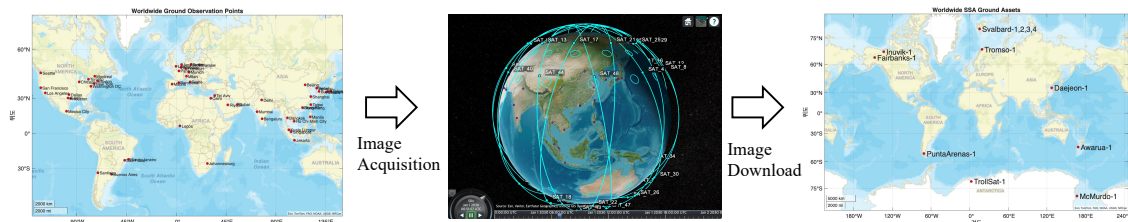


Fig. 26: End-to-end Earth-observation mission sequence from image acquisition to image download

For each imaging event, the performance analysis searches the subsequent ground-station opportunities available to the spacecraft and records whether the image can be downloaded within the analysis horizon. The resulting metric is the imaging-to-downlink latency, which complements the image-capture revisit metric by accounting for the time required to return data to the ground segment. This distinction is important because a constellation may observe a target frequently while still producing long data-delivery delays if the downlink network is sparse or heavily constrained.

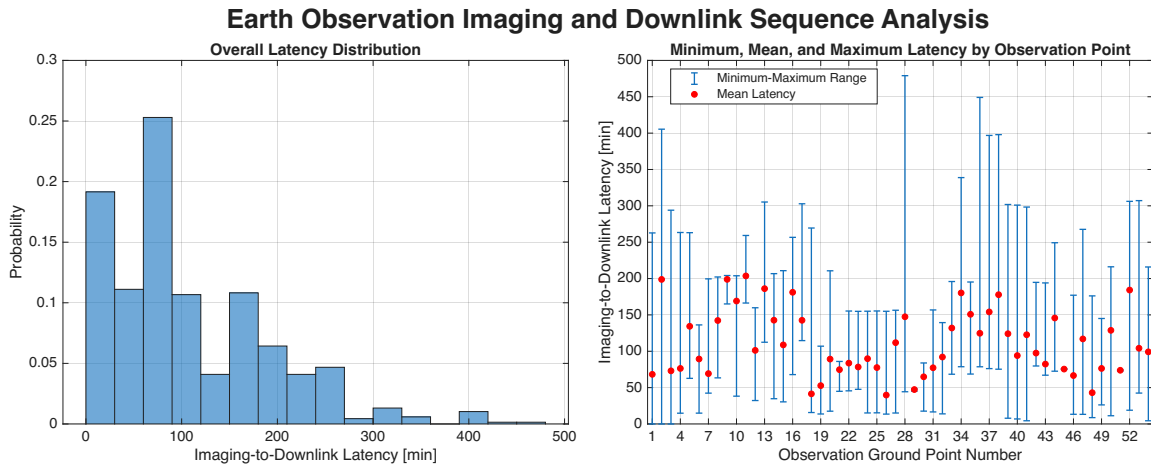


Fig. 27: EO imaging and downlink performance, including the overall imaging-to-downlink latency distribution and ground-point-specific latency statistics

Table 10: Imaging-to-downlink performance analysis.

Performance Metric	Result
Number of Satellites	48
Operation Cadence for Each Satellite (τ_{sat})	1000 s
Number of Imaging Ground Points	54
Number of Downlink Ground Stations	12
Operation Cadence for Each Ground Station (τ_{GS})	1500 s
Total Imaging Events	718
Successfully Downlinked Events	684
Failed Downlink Events	34
Downlink Success Rate	95.26%
Mean Imaging-to-Downlink Latency	108.77 min
Median Imaging-to-Downlink Latency	85.21 min
Maximum Imaging-to-Downlink Latency	479.00 min

Figure 27 summarizes the end-to-end result in two complementary views. The latency histogram is right-skewed, with most successful events delivered considerably earlier than the longest-tail cases and the ground-point-specific min-max plots reveal that a smaller subset of locations experiences substantially larger worst-case latency. These plots show that a single network-average metric can hide geographically dependent delivery risk.

Table 10 reports 718 imaging events, of which 684 are successfully downlinked and 34 remain unsuccessful within the analyzed sequence, corresponding to a 95.26% success rate. Among the successful events, the mean latency is 108.77 min, the median is 85.21 min, and the maximum reaches 479.00 min. The difference between the mean and median is consistent with the long right tail visible in Figure 27. Consequently, future scheduling formulations should consider image-acquisition revisit and downlink latency jointly when the mission objective is rapid delivery rather than observation opportunity alone.

5. CONCLUSION AND FUTURE WORK

This paper presented a generalized contact-selection framework for task scheduling in LEO satellite constellations. The central idea is to separate geometric opportunity generation from operational scheduling: orbit propagation first produces an unconstrained time-ordered contact chart, and a mixed-integer optimization then selects the subset of contacts that can be executed under resource-specific cadence constraints. Selection and difference matrices provide a common algebraic representation for both the local operation constraints and the target-side revisit-time objective.

The resulting structure applies naturally in both SAT-to-GS and GS-to-SAT directions. In SAT-to-GS missions, satellite payload cadence is constrained while ground-point revisit or communication period is optimized. In GS-to-SAT missions, ground-resource cadence is constrained while satellite observation or service gaps are optimized. The case studies demonstrated that the same model can represent worldwide Earth-observation imaging, weighted priority service, simplified 6G multi-beam gateway scheduling, distributed SSA ground-asset scheduling, colocated high-demand ground resources, and end-to-end image-data download analysis.

The numerical studies also identify several operational trade-offs. Increasing τ reduces the number of executable contacts and degrades revisit performance, directly quantifying the cost of more restrictive resource cadence. Increasing selected diagonal weights in $\mathbf{G}$ can improve service for a priority location without rebuilding the underlying MILP. Adding independent payload or ground-asset channels increases parallel service capacity and can recover a significant fraction of the performance lost to single-resource cadence constraints.

Several extensions are required before the framework can represent a complete operational system. Future work will incorporate access duration in addition to point-contact approximations, attitude-transition time, energy and onboard-storage constraints, heterogeneous payload duty cycles, and explicit data-volume requirements. For communications missions, link capacity, beam interference, frequency assignment, queueing, and ISL routing can be introduced as additional coupled constraints. For SSA and Earth observation, uncertainty in demand, weather, sensor availability, and orbit knowledge can be handled through adaptive or robust scheduling. Finally, asynchronous distributed optimization and multi-agent learning approaches are promising directions for moving from pre-planned scheduling toward onboard or ground-network autonomy under dynamically changing mission requests.

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